

Unknown angles in triangles and quadrilaterals

180° in a triangle, 360° in a quadrilateral, and the special figures that follow.

LESSON 84–86

3 × 40 MIN

MATEMATIKA 6, §17

S7 5.2

LESSON CLOCK

25'

30'

35'

25'

5

The angle sum of a triangle	25 min
Special triangles	30 min
Quadrilaterals	35 min
Figures made of triangles	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the angle sum of a triangle and of a quadrilateral.
- Use the base angles of an isosceles triangle.
- Find angles in figures made of several triangles.
- Write an equation for an unknown angle in a figure.

TEXTBOOK

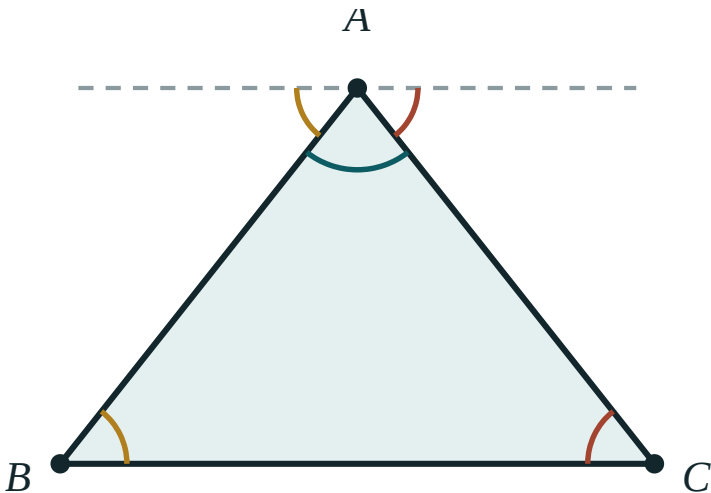
National	pp. 237–246
Cambridge	Stage 7, pp. 58–62
Workbook	Exercise 5.3

01

Explanation

The angle sum of a triangle

$$\angle A + \angle B + \angle C = 180^\circ$$



the three angles fill 180°

The three angles of a triangle

GIVEN	WORKING	THIRD ANGLE
$50^\circ, 60^\circ$	$180 - 110$	70°
$90^\circ, 35^\circ$	$180 - 125$	55°
$120^\circ, 25^\circ$	$180 - 145$	35°
angles $x, 2x, 3x$	$6x = 180$	$30^\circ, 60^\circ, 90^\circ$

TEAR THE CORNERS OFF AND CHECK

The three corners of any paper triangle lay flat along a straight line. It is not a proof, but it is convincing, and the proof waits until Grade 7.

Special triangles

KIND	WHAT IS EQUAL	ANGLES
equilateral	all three sides	$60^\circ, 60^\circ, 60^\circ$
isosceles	two sides	the two base angles are equal
right-angled	one angle is 90°	the other two add to 90°
right isosceles	two sides and a right angle	$90^\circ, 45^\circ, 45^\circ$

ISOSCELES TRIANGLE	WORKING	ANSWER
apex 40°	$(180 - 40) \div 2$	70° each
base angle 50°	$180 - 100$	80°

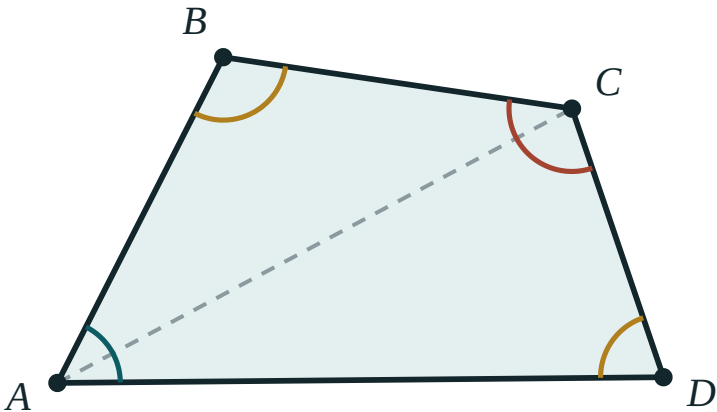
READ WHICH ANGLE IS GIVEN

An apex of 40° gives base angles of 70° ; a base angle of 40° gives an apex of 100° . The same number, two different triangles.

Quadrilaterals

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

A diagonal cuts any quadrilateral into two triangles, so its angles add to $2 \cdot 180^\circ$.



$$2 \times 180^\circ = 360^\circ$$

A quadrilateral split into two triangles

FIGURE	ANGLE FACTS
any quadrilateral	the four add to 360°
rectangle	four right angles
parallelogram	opposite angles equal; neighbours add to 180°
rhombus	as a parallelogram, and all sides equal
kite	one pair of opposite angles equal

TWO TRIANGLES, ONE QUADRILATERAL

That single picture explains the 360° , and it generalises: a pentagon splits into three triangles and has an angle sum of 540° .

Figures made of triangles

PROBLEM	ROUTE	ANSWER
a triangle with angles $x, x, 80^\circ$	$2x = 100$	50°
a quadrilateral with $90^\circ, 90^\circ, 130^\circ$	$360 - 310$	50°
a parallelogram with one angle 72°	neighbours add to 180°	108°
an exterior angle of a triangle, interior 115°	$180 - 115$	65°

WORK OUTWARDS FROM WHAT YOU KNOW

Fill in every angle you can, even ones the question did not ask for; the wanted angle usually appears after two or three of them are written on the diagram.

02 Worked examples

EXAMPLE 1 A triangle has angles 50° and 60° . Find the third.

1 $50 + 60 = 110$

2 $180 - 110 = 70$

3 70°

ANSWER

70°

EXAMPLE 2 An isosceles triangle has an apex angle of 40° . Find the base angles.

- ① The two base angles are equal.

- ② $180 - 40 = 140$ shared between them.

- ③ $140 \div 2 = 70^\circ$ each.

ANSWER

70° each

EXAMPLE 3 A quadrilateral has angles 90° , 90° and 130° . Find the fourth.

- ① The four add to 360° .

- ② $90 + 90 + 130 = 310$

- ③ $360 - 310 = 50^\circ$

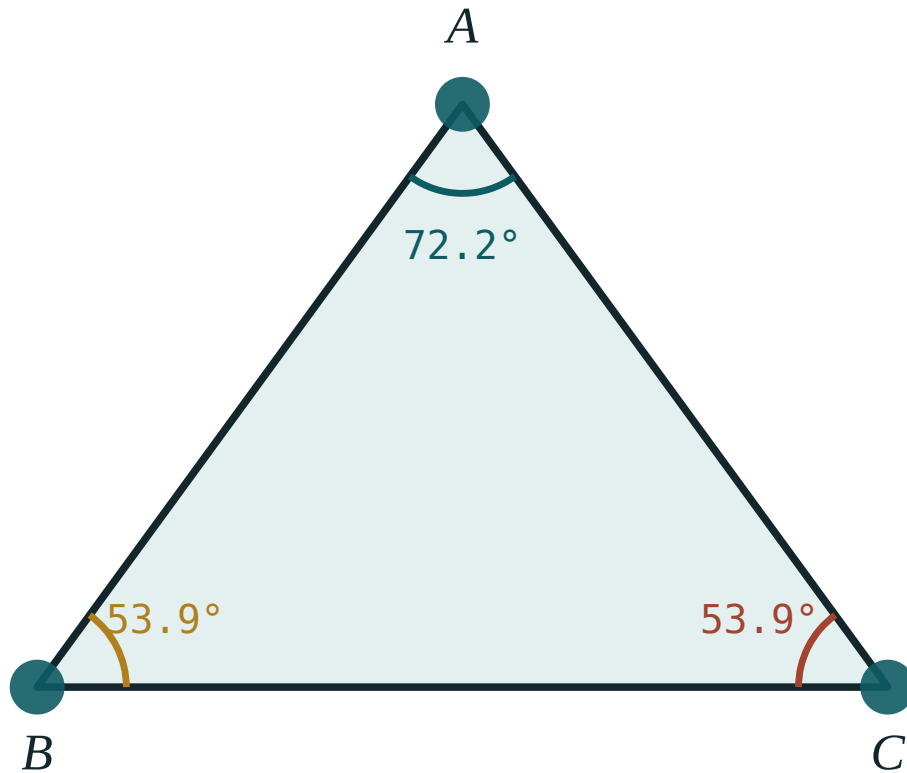
ANSWER

50°

03 Interactive model

Give a quadrilateral and ask the class to draw one diagonal; the 360° is then discovered rather than announced.

Drag a vertex — the sum stays at 180°



$\angle A$ 72.2°

$\angle B$ 53.9°

$\angle C$ 53.9°

sum 180°

longest side **BC**

largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Triangle	Uchburchak	Треугольник
Quadrilateral	To'rtburchak	Четырёхугольник
Angle sum	Burchaklar yig'indisi	Сумма углов
Isosceles	Teng yonli	Равнобедренный
Equilateral	Teng tomonli	Равносторонний
Base angles	Asosdagi burchaklar	Углы при основании
Parallelogram	Parallelogramm	Параллелограмм
Diagonal	Diagonal	Диагональ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angles of a triangle add to:

The angles of a quadrilateral add to:

An equilateral triangle has angles of:

An isosceles triangle with apex 40° has base angles:

A parallelogram with one angle 72° has a neighbour of:

A pentagon's angles add to:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 50° and 60° : the third angle

ANSWER 70°

2. 90° and 35°

ANSWER 55°

3. 120° and 25°

ANSWER 35°

4. Each angle of an equilateral triangle

ANSWER 60°

5. Isosceles with apex 40° : the base angles

ANSWER 70°

6. Isosceles with base angle 50° : the apex

ANSWER 80°

7. A quadrilateral with 90° , 90° , 130°

ANSWER 50°

Medium · the standard question

7 PROBLEMS

1. A triangle with angles x , $2x$, $3x$

ANSWER $30^\circ, 60^\circ, 90^\circ$

2. A triangle with x , x , 80°

ANSWER 50° each

3. A parallelogram with one angle 72°

ANSWER 108°

4. A right isosceles triangle

ANSWER $90^\circ, 45^\circ, 45^\circ$

5. A quadrilateral with x , x , $2x$, $2x$

ANSWER $60^\circ, 60^\circ, 120^\circ, 120^\circ$

6. An exterior angle beside an interior angle of 115°

ANSWER 65°

7. The angle sum of a pentagon

ANSWER 540°

Hard · stretch and reason

7 PROBLEMS

1. A triangle with angles in the ratio $2 : 3 : 4$

ANSWER $40^\circ, 60^\circ, 80^\circ$

2. A quadrilateral with angles in the ratio $1 : 2 : 3 : 4$

ANSWER $36^\circ, 72^\circ, 108^\circ, 144^\circ$

3. An isosceles triangle with one angle 100° : the others

ANSWER 40° each

4. An isosceles triangle with one angle 40° : two possibilities

ANSWER $40^\circ, 70^\circ, 70^\circ$ or $40^\circ, 40^\circ, 100^\circ$

5. A kite with angles $110^\circ, 80^\circ, 80^\circ$: the fourth

ANSWER 90°

6. Why do a quadrilateral's angles add to 360° ?

ANSWER A diagonal splits it into two triangles

7. The angle sum of a hexagon

ANSWER 720°

07 Homework

Homework — 5 tasks

Fill in every angle you can find, not only the one asked for.

1. A triangle has angles 72° and 43° . Find the third.
2. An isosceles triangle has an apex angle of 34° . Find the base angles.
3. A quadrilateral has angles $85^\circ, 95^\circ$ and 110° . Find the fourth.
4. A triangle has angles in the ratio $3 : 4 : 5$. Find them.
5. A parallelogram has one angle of 115° . Find all four.